7(a) ${}^{90}_{38}Sr \rightarrow {}^{90}_{39}Y + {}^{0}_{-1}e$ [1; no need neutrino]

(b) Energy released = (mass defect) c^2

$= (m_{Sr} - m_Y - m_e)\, c^2$ [1]

$= [(89.907738 - 89.907151) \times 1.66 \times 10^{-27} - 9.11 \times 10^{-31}] \times (3.00 \times 10^8)^2$

$= 5.7078 \times 10^{-15}$ [1]

$= 5.71 \times 10^{-15}$ J

(c)(i) Total energy that needs to be released per second, $E = \dfrac{\text{Power to be supplied}}{\text{Efficiency}}$

$= \dfrac{155}{0.070}$

$= 2214.3$ J [1]

Activity $= \dfrac{E}{\text{Energy released in one reaction}}$

$= \dfrac{2214.3}{5.71 \times 10^{-15}}$

$= 3.8779 \times 10^{17}$ [1]

$= 3.88 \times 10^{17}$ Bq

(ii) $A = \lambda N$

Number of strontium-90 needed $N = \dfrac{A}{\lambda}$ [1]

$= \dfrac{A\, \tau_{1/2}}{\ln 2}$

$= \dfrac{3.8779 \times 10^{17} \times 29 \times 365 \times 24 \times 3600}{\ln 2}$

$= 5.1165 \times 10^{26}$

$= 5.12 \times 10^{26}$ [1]

(iii) Mass of strontium-90 needed = number of nuclei x mass of 1 nuceus [1]

$= 5.1165 \times 10^{26} \times 89.907738 \times 1.66 \times 10^{-27}$

$= 76.4$ kg [1]

8(a)(i)